

Business Insights Report

Project Title

Store Sales Analytics Dashboard

1. Executive Summary

This project analyzes historical store sales data from the Kaggle Store Sales Forecasting dataset. The objective is to clean, process, analyze, and visualize sales data to identify trends, understand store performance, evaluate promotional effectiveness, and generate actionable business insights.

An interactive dashboard was developed using Streamlit and Plotly to allow users to explore sales performance through multiple filters and visualizations.

2. Project Objectives

The objectives of this project were:

- Perform data cleaning and preprocessing.
 - Merge multiple datasets into one master dataset.
 - Conduct Exploratory Data Analysis (EDA).
 - Build an interactive dashboard.
 - Generate business insights from historical sales data.
 - Present findings using visualizations.
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3. Dataset Description

Dataset Source: Kaggle - Store Sales Forecasting

Files Used:

- train.csv
- stores.csv
- transactions.csv
- oil.csv
- holidays_events.csv

The dataset contains:

- Daily sales records
- Product categories
- Store information

- Oil prices
 - Holiday information
 - Customer transactions
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4. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
 - Handled missing values.
 - Converted date columns into datetime format.
 - Standardized categorical columns.
 - Merged all datasets into a single master dataset.
 - Created additional time-based features.
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5. Exploratory Data Analysis

EDA included:

- Daily sales trend analysis.
 - Monthly sales comparison.
 - Product category analysis.
 - Store-wise sales comparison.
 - City-wise sales comparison.
 - Promotion impact analysis.
 - Correlation analysis.
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6. Dashboard Features

The dashboard provides:

- KPI cards
 - Interactive filters
 - Daily sales trend
 - Monthly sales chart
 - Product category distribution
 - Top performing stores
 - Sales by city
 - Promotion analysis
 - Correlation heatmap
 - Download filtered dataset
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7. Key Business Insights

Insight 1

Grocery products generated the highest overall sales among all product categories.

Insight 2

Promotional campaigns positively influenced total sales.

Insight 3

Sales performance varies significantly across cities and stores.

Insight 4

Sales exhibit seasonal and monthly trends.

Insight 5

Higher customer transactions generally correspond to higher sales.

8. Conclusion

The Store Sales Analytics Dashboard provides valuable business intelligence by transforming raw sales data into meaningful insights.

The dashboard enables business users to monitor sales performance, compare stores, analyze product categories, evaluate promotional campaigns, and support data-driven decision-making.

Future enhancements may include machine learning-based sales forecasting and real-time dashboard integration.